

Geometry on Statistical Models

Math Club Talk

Soham Bakshi

Indian Statistical Institute

October 10, 2021

Overview

Riemannian Geometry

Parametric Models and Fisher Information

Rao-Fisher Geometry

α -connections

Dual Connections

What's the idea?

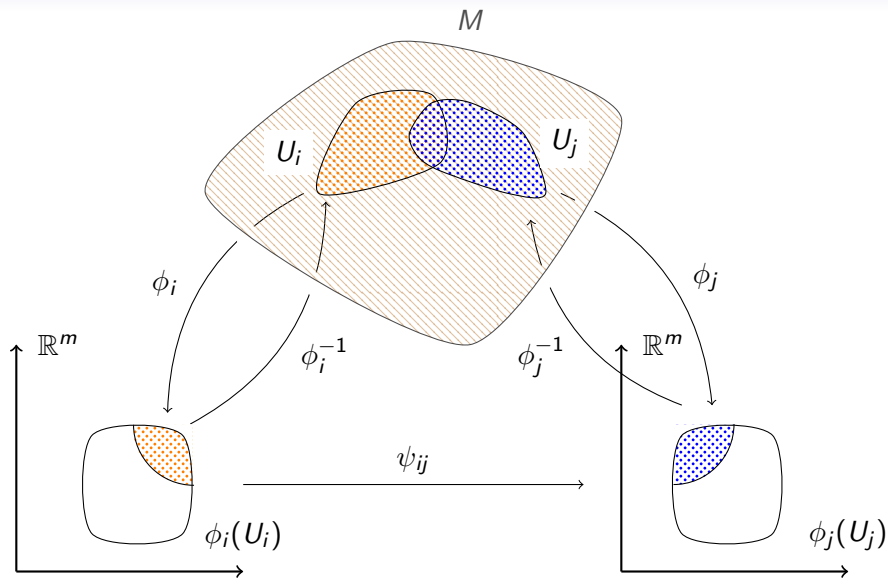
Finding geometric structures in the space or subspaces of probability distributions.

1. **Distance, Divergence:** How far apart two distributions are?
What can be a measure of difference between two populations?
2. **Information Loss:** How can we geometrically interpret information loss? How can we go from one distribution to other by losing least information? (*smallest information transform*)
3. **Statistical Curvature:** Exponential families are well behaved. How can one quantify departure from it? *Efron (1975)*

Manifolds

Manifolds are patches of locally euclidean spaces glued together.

We call them *smooth* if the transition maps $\psi_{ij} = \phi_j \circ \phi_i^{-1}$ are smooth.



Tangent Space

The set of all tangent vectors to M at p is the **tangent space** $(T_p M)$

Tangent space at a point $p \in M$ can be intuitively contains all possible directions passing “*tangentially*” through it.

The tangent space at any point of a n dimensional smooth manifold is a n -dimensional vector space. If a chart $\varphi = (x^1, \dots, x^n) : U \rightarrow \mathbb{R}^n$ is given with $p \in U$, then $\left\{ \left(\frac{\partial}{\partial x^1} \right)_p, \dots, \left(\frac{\partial}{\partial x^n} \right)_p \right\}$ forms a basis of $T_p M$.

Riemannian Metric

Let M be a smooth manifold. A **Riemannian metric** g on M is a *smooth* family of inner products on the tangent spaces of M . Namely, g associates to each $p \in M$ a **positive definite symmetric bilinear form** on $T_p M$,

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R}$$

and the smoothness condition on g refers to the fact that the function

$$p \in M \rightarrow g_p(X_p, Y_p) \in \mathbb{R}$$

is smooth.

Vector Fields

A **vector field** X on a smooth manifold M is a correspondence that associates to each point $p \in M$ a vector $X(p) \in T_p M$. In terms of mappings, X is a *mapping of M into the tangent bundle TM* . The field is smooth if the mapping $X : M \rightarrow TM$ is smooth.

$\mathcal{X}(M)$ denotes the set of all vector fields $X : M \rightarrow TM$.

Affine Connections

An **affine (or linear) connection** ∇ on a smooth manifold M is a mapping

$$\nabla : \mathcal{X}(M) \times \mathcal{X}(M) \rightarrow \mathcal{X}(M)$$

which is denoted by $(X, Y) \xrightarrow{\nabla} \nabla_X Y$ and which satisfies the following properties :

1. $\nabla_{fX+gY}Z = f\nabla_X Z + g\nabla_Y Z$
2. $\nabla_X(Y + Z) = \nabla_X Y + \nabla_X Z$
3. $\nabla_X(fY) = f\nabla_X Y + X(f)Y$

in which $X, Y, Z \in \mathcal{X}(M)$ and $f, g \in \mathcal{D}(M)$.

Affine Connections

An **affine (or linear) connection** ∇ on a smooth manifold M is a mapping

$$\nabla : \mathcal{X}(M) \times \mathcal{X}(M) \rightarrow \mathcal{X}(M)$$

which is denoted by $(X, Y) \xrightarrow{\nabla} \nabla_X Y$ and which satisfies the following properties :

1. $\nabla_{fX+gY}Z = f\nabla_X Z + g\nabla_Y Z$
2. $\nabla_X(Y + Z) = \nabla_X Y + \nabla_X Z$
3. $\nabla_X(fY) = f\nabla_X Y + X(f)Y$

in which $X, Y, Z \in \mathcal{X}(M)$ and $f, g \in \mathcal{D}(M)$.

For any $V, W \in \mathcal{X}(M)$, $\nabla_W V$ is called the *covariant derivative* of V along W .

Levi-Civita Connection

Theorem

(*Fundamental Lemma of Riemannian Geometry*) Let (M, g) be a Riemannian manifold. There exists a unique linear connection ∇ on M that is compatible with g and symmetric.

A linear connection ∇ is said to be compatible with g if it satisfies the following product rule for all vector fields X, Y, Z .

$$\nabla_X g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

.

Levi-Civita Connection

Theorem

(*Fundamental Lemma of Riemannian Geometry*) Let (M, g) be a Riemannian manifold. There exists a unique linear connection ∇ on M that is compatible with g and symmetric.

A linear connection ∇ is said to be compatible with g if it satisfies the following product rule for all vector fields X, Y, Z .

$$\nabla_X g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

.

The connection is called **Levi-Civita connection** with respect to g .

Christofel Symbols

By prescribing smooth functions $\Gamma_{ij,k}(p)$ or $\Gamma_{ij}^k(p)$, called the **Christoffel symbols** of the first and second kind respectively, one can define the unique affine connection ∇ that satisfies (locally)

$$g(\nabla_{e_i} e_j, e_k) = \Gamma_{ij,k}, \quad \nabla_{e_i} e_j = \sum_k \Gamma_{ij}^k e_k$$

However, the values of $\Gamma_{ij}^k(p)$ and $\Gamma_{ij,k}(p)$ are different for different coordinate systems, the former shows how tangent vectors changes on a manifold, thus shows how basis vectors changes.

Flatness

$(x^1, \dots, x^n)$ be an **affine coordinate system** for the connection ∇ if the n basis vector fields $e_i = \partial_i = \frac{\partial}{\partial x^i}$ are all parallel on S .
Equivalent conditions for a coordinate system to be an affine coordinate system:

$$\begin{aligned}\nabla_{\partial_i} \partial_j &= \sum_{k=1}^n \left(\Gamma_{ij}^k \partial_k \right) = 0 \text{ for all } i \text{ and } j \\ \Gamma_{ij}^k &= 0 \quad \text{for all } i, j \text{ and } k\end{aligned}$$

A manifold is **flat** w.r.t the connection ∇ if there exists an affine coordinate system for the connection ∇ .

Geodesics

Let M be a manifold with a linear connection ∇ , and let γ be a curve in M .

The **acceleration** of γ is the vector field $D_t \dot{\gamma}$ along γ .

A curve γ is called a **geodesic** with respect to ∇ if its **acceleration is zero**: $D_t \dot{\gamma} \equiv 0$.

Geodesics

Let M be a manifold with a linear connection ∇ , and let γ be a curve in M .

The **acceleration** of γ is the vector field $D_t\dot{\gamma}$ along γ .

A curve γ is called a **geodesic** with respect to ∇ if its **acceleration is zero**: $D_t\dot{\gamma} \equiv 0$.

Example: The geodesics on $\mathbb{R}^n$ are exactly the straight lines with constant speed parametrizations.

Curve Length and Distance

If $\gamma : [a, b] \rightarrow M$ is a curve segment, we define the **length** of γ to be

$$L(\gamma) := \int_a^b \|\dot{\gamma}(t)\| dt = \int_a^b \sqrt{g_{ij} \dot{\gamma}^i \dot{\gamma}^j} dt = \int_a^b \sqrt{\dot{\gamma}(g_{ij}) \dot{\gamma}'} dt$$

For any pair of points $p, q \in M$, we define the **Riemannian distance** $d(p, q)$ to be the *infimum of the lengths of all curves from p to q* . If $d(p, q) = L(\gamma)$ for some γ joining p to q , then γ is called a **minimizing curve**.

Two Results

Theorem

Every minimizing curve is a geodesic when given a unit speed parametrization.

Two Results

Theorem

Every minimizing curve is a geodesic when given a unit speed parametrization.

Theorem

Every Riemannian geodesic is locally minimizing.

RG Set-Up

For Riemannian geometric modelling:

1. Start with a smooth manifold.
2. Define Riemannian metric g_p . Equivalently define $g_p(e_i, e_j) = g_p(\partial_i, \partial_j)$ for all i, j .
3. Define affine connection (or specify Christofel symbols). Only Levi-Civita?

Parametric Models

A **parametric statistical model** $\mathcal{S}$ is a family of probability density functions $p(x; \theta)$ uniquely determined by parameters $\theta = (\theta_1, \dots, \theta_n) \in \Theta$.

$$\mathcal{S} = \{p_\theta = p(x; \theta) \mid \theta = (\theta^1, \dots, \theta^n) \in \Theta\}$$

The set $\Theta \subset \mathbb{R}^n$ is called the parameters space. The set $\mathcal{S}$ is a subset of the infinite dimensional space of L^1 functions

$$\mathcal{P}(\mathcal{X}) = \left\{ f; f: \mathcal{X} \rightarrow \mathbb{R}, f \geq 0, \int_{\mathcal{X}} f = 1 \right\}$$

Regularity Conditions

A statistical model S is **geometrically regular** if it satisfies the following regularity conditions

1. Θ is homeomorphic to an open subset of $\mathbb{R}^n$.
 2. Topology of S induced from $\mathbb{R}^n$ is compatible with subspace topology of S in L^1 space.
 3. Support of $p(x; \theta)$ does not depend on θ .
 4. $\frac{\partial p_\theta}{\partial \theta^1}, \dots, \frac{\partial p_\theta}{\partial \theta^n}$ are linearly independent.
- ... and some exchangeability and smoothness conditions.

Score Functions

A useful mapping is the log-likelihood function $\ell : \mathcal{S} \rightarrow \mathcal{F}(\mathcal{X}, \mathbb{R})$ defined by

$$\ell(p_{\theta})(x) = \ln p_{\theta}(x)$$

Sometimes, for convenience reasons, this will be denoted by $\ell_x(\theta) = \ell(p(x))$. The derivatives of the log-likelihood function are called **score functions**

$$\partial_j \ell_x(\theta) = \frac{\partial \ell_x(\theta)}{\partial \theta^j}, \quad 1 \leq j \leq n$$

Heuristically, they describe how the information contained in p_{θ} changes in the direction of θ^j . The functions $\partial_j \ell_x(\theta)$ will play a central role in defining the Fisher information matrix and the entropy on a statistical manifold.

Information Matrix

Given parameter $\boldsymbol{\theta} = (\theta_1, \dots, \theta_n) \in \Theta$ and the log-likelihood function $\ell(\xi)$. The **Fisher information matrix** is defined by

$$I_{ij}(\boldsymbol{\theta}) = E_{\boldsymbol{\theta}} [\partial_i \ell(\boldsymbol{\theta}) \partial_j \ell(\boldsymbol{\theta})], \quad \forall i, j \in \{1, \dots, n\}$$

which can be written explicitly as

$$I_{ij}(\boldsymbol{\theta}) = \int_{\mathcal{X}} \partial_{\theta_i} \ln p_{\boldsymbol{\theta}}(x) \cdot \partial_{\theta_j} \ln p_{\boldsymbol{\theta}}(x) \cdot p_{\boldsymbol{\theta}}(x) dx$$

We shall assume that the integral exists. Equivalently,

$$I_{ij}(\boldsymbol{\theta}) = 4 \int_{\mathcal{X}} \partial_{\theta_i} \sqrt{p_{\boldsymbol{\theta}}(x)} \cdot \partial_{\theta_j} \sqrt{p_{\boldsymbol{\theta}}(x)} dx.$$

Theorem

*The Fisher information matrix on any geometrically regular statistical model is **symmetric** and **positive definite**.*

Proof.

The symmetry follows from formula. For all θ , $\forall \mathbf{v} \neq 0$, we find

$$\begin{aligned}\mathbf{v} G \mathbf{v}' &= \sum_{i,j} I_{ij} v^i v^j = 4 \sum_{i,j} \left(\int_{\mathcal{X}} v^i \partial_{\theta^i} \sqrt{p_{\theta}(x)} \cdot v^j \partial_{\theta^j} \sqrt{p_{\theta}(x)} \right) dx \\ &= 4 \int_{\mathcal{X}} \left(\sum_i v^i \partial_{\theta^i} \sqrt{p} \right) \left(\sum_j v^j \partial_{\theta^j} \sqrt{p} \right) dx \\ &= 4 \int_{\mathcal{X}} \left(\sum_i v^i \partial_{\theta^i} \sqrt{p} \right)^2 dx \geq 0\end{aligned}$$

so (I_{ij}) is non-negative definite. Furthermore, as $\{\partial_{\theta^i} p\}$ are independent (I_{ij}) is positive definite.



Calculating I_{ij}

There are various equivalent formulations of Fisher information number and matrix. We shall use the following: *The Fisher information matrix can be written as the negative expectation of the Hessian of the log-likelihood function*

$$I_{ij}(\theta) = -E_{\theta} [\partial_{\theta^i} \partial_{\theta^j} \ell(\theta)] = -E_{\theta} [\partial_{\theta^i} \partial_{\theta^j} \ln p_{\theta}]$$

Exponential Distribution

In this case $p_\theta(x) = \theta e^{-\theta x}$ and then $\partial_\theta^2 \ln p_\theta = -\frac{1}{\theta^2}$. Therefore

$$I = I_{11} = -E \left[\partial_\theta^2 \ln p_\theta \right] = \int_0^\infty \frac{1}{\theta^2} p_\theta(x) dx = \frac{1}{\theta^2}$$

The length of a curve $\theta = \theta(t)$ joining the points $\theta_0 = \theta(0)$ and $\theta_1 = \theta(1)$ is

$$\int_0^1 \sqrt{g(\dot{\theta}(t), \dot{\theta}(t))} dt = \int_0^1 \sqrt{g_{11} \dot{\theta}^2(t)} dt = \int_0^1 \left| \frac{\dot{\theta}(t)}{\theta(t)} \right| dt = \left| \ln \frac{\theta_1}{\theta_0} \right|$$

This induces on the one-dimensional statistical model $\mathcal{X} = (0, \infty)$ the following hyperbolic distance function

$$d(x_1, x_2) = |\ln x_2 - \ln x_1|$$

Exponential Family

The exponential family is

$$p(x; \theta) = e^{C(x) + \theta^i F_i(x) - \psi(\theta)}$$

We can represent the Fisher information matrix in terms of the functions $F_i(x)$, or in terms of ψ

$$I_{ij}(\theta) = E_{\theta} [\partial_i \ell_x(\theta) \partial_j \ell_x(\theta)] = \text{Cov}_{\theta} (F_i, F_j)$$

$$I_{ij}(\theta) = -E_{\theta} [\partial_i \partial_j \ell_x(\theta)] = E_{\theta} [\partial_i \partial_j \psi(\theta)] = \partial_i \partial_j \psi(\theta)$$

It is worthy to note the role of convexity of function ψ in the positive definiteness of g_{ij} .

Mixture Family

The probability density family

$$p(x; \boldsymbol{\theta}) = C(x) + \theta^i F_i(x)$$

We obtain

$$l_{ij}(\boldsymbol{\theta}) = -E_{\boldsymbol{\theta}} [\partial_i \partial_j \ell_x(\boldsymbol{\theta})] = E_{\boldsymbol{\theta}} \left[\frac{F_i(x) F_j(x)}{p(x; \boldsymbol{\theta})^2} \right] = \int_{\mathcal{X}} \frac{F_i(x) F_j(x)}{p(x; \boldsymbol{\theta})} dx$$

The “Pathbreaking” Paper

Rao, C. R. (1945), *Information and the accuracy attainable in the estimation of statistical parameters*, Bull. Calcutta Math. Soc. 37, 81–89.

- **The Cramér-Rao lower bound:** How good can a statistical estimator be? Is there a fundamental limit when estimating statistical parameters?
- **Rao-Fisher Distance:** First introduced differential geometric approach on statistical manifolds, gave birth to *information geometry*.

The Idea

Quadratic differential form (the infinitesimal element lengths):

$$\begin{aligned} ds^2(\theta) &= I_{ij}(\theta) d\theta_i d\theta_j \\ &= (\nabla\theta)^\top I(\theta) \nabla\theta \end{aligned}$$

And, the distance as:

$$D(p_{\theta_1}(x), p_{\theta_2}(x)) = \min_{\substack{\theta(t) \\ \theta(0)=\theta_1, \theta(1)=\theta_2}} \int_0^1 \left(\sqrt{(\nabla\theta)^\top I(\theta) \nabla\theta} \right) dt$$

Basically, the geometric modelling is

$$g_p(\partial_i, \partial_j) = g_{ij}(p) = I_{ij}(p)$$

Computing Fisher-Rao Distance

To compute Rao's distance we need to calculate explicitly the geodesic linking $p_{\theta_1}(x)$ to $p_{\theta_2}(x)$. This is done by solving the following second order ordinary differential equation (ODE) :

$$g_{ki}\ddot{\theta}_i + \Gamma_{k,ij}\dot{\theta}_i\dot{\theta}_j = 0.$$

The coefficients $\Gamma_{k,ij}$ are the Christoffel symbols of the first kind defined by:

$$\Gamma_{k,ij} = \frac{1}{2} \left(\frac{\partial g_{ik}}{\partial \theta_j} + \frac{\partial g_{kj}}{\partial \theta_i} - \frac{\partial g_{ij}}{\partial \theta_k} \right).$$

In practice, it is difficult to explicitly compute the geodesics of the Fisher-Rao geometry of arbitrary models, and one needs to perform a gradient descent to find a local solution for the geodesics.

Why is it a good choice?

- (I_{ij}) is symmetric and positive definite, so suitable for Riemannian metric
- (I_{ij}) is invariant under reparametrizations of the sample space.
- (I_{ij}) is covariant under reparametrizations of the parameters space Θ (i.e., transforms as a 2-covariant tensor).
- Setting $g_{ij} = I_{ij}$ gives the **unique** riemannian metric satisfying the above two invariance properties.
- Let $I(\theta)$ be the Fisher information matrix of $S = p(x; \theta)$, and $I_F(\theta)$ be the Fisher information matrix of the induced model $S_F = q(y; \theta)$, then we have $I_F(\theta) \leq I(\theta)$ in the sense that $\Delta I(\theta) = I_F(\theta) - I(\theta)$ is positive semidefinite. $\Delta I(\theta) = 0$ iff F is a sufficient statistic for S .

Other Statistical Distances

- **Mahalanobis Distance:** The Mahalanobis D -squared distance of a vector x to a group of vectors with covariance matrix Σ and mean μ is $D_{\Sigma}^2(x, \mu) = (x - \mu)\Sigma^{-1}(x - \mu)'$. The generic Mahalanobis distance

$$D_M(p, q) = \sqrt{(p - q)^{\top} M (p - q)}$$

(with M positive definite)

- **Bhattacharyya Distance:** between two distributions indexed by parameters θ_1 and θ_2 is defined by

$$B(\theta_1, \theta_2) = -\log \int_{x \in \mathcal{X}} \sqrt{p_{\theta_1}(x)p_{\theta_2}(x)} dx$$

α -connections

Let $S = \{p_{\theta}\}$ be an n -dimensional model, and consider the function $\Gamma_{ij,k}^{(\alpha)}$ which maps each point θ to the following value:

$$\left(\Gamma_{ij,k}^{(\alpha)}\right)_{\theta} = E_{\theta} \left[\left(\partial_i \partial_j \ell_{\theta} + \frac{1-\alpha}{2} \partial_i \ell_{\theta} \partial_j \ell_{\theta} \right) (\partial_k \ell_{\theta}) \right]$$

where α is an arbitrary real number. We defined an affine connection $\nabla^{(\alpha)}$ which satisfy:

$$g \left(\nabla_{\partial_i}^{(\alpha)} \partial_j, \partial_k \right) = \Gamma_{ij,k}^{(\alpha)}$$

where g is the Fisher metric. We call $\nabla^{(\alpha)}$ the α -connection.

Properties

- α -connections are **symmetric**.
- Relationship between α -connection and β -connection:

$$\Gamma_{ij,k}^{(\beta)} = \Gamma_{ij,k}^{(\alpha)} + \frac{\alpha - \beta}{2} E [\partial_i \ell_{\theta} \partial_j \ell_{\theta} \partial_k \ell_{\theta}]$$

- The 0 -connection is the *Levi-Civita connection* with respect to the Fisher metric.

-

$$\nabla^{(\alpha)} = \frac{1 + \alpha}{2} \nabla^{(1)} + \frac{1 - \alpha}{2} \nabla^{(-1)}$$

Exponential and Mixture Connections

- Exponential family is $\nabla^{(1)}$ -flat. The natural parameters form a (1)-affine coordinate system.
- Mixture family is $\nabla^{(-1)}$ -flat. The natural parameters form a (-1)-affine coordinate system.

Dual Connections

Let (M, g, ∇, ∇^*) be a Riemannian manifold with metric g and two affine connections ∇ and ∇^* . If for all vector fields $X, Y, Z \in \mathcal{X}(S)$

$$Zg(X, Y) = g(\nabla_Z X, Y) + (X, \nabla_Z^* Y)$$

hold, we say that ∇ and ∇^* are **duals** of each other w.r.t. g and call one the dual connection of the other. Additional, we call the triple (g, ∇, ∇^*) a **dualistic structure** on S . If M is both ∇ -flat and ∇^* -flat, we call M **dually flat**.

A Key Result

Theorem

For statistical models with Rao-Fisher geometry, $\nabla^{(\alpha)}$ -flat iff $\nabla^{(-\alpha)}$ -flat, this is true for any α .

Exponential and Mixture families are dually flat w.r.t $(g, \nabla^1, \nabla^{-1})$.

Abstract Statistical Manifolds

A **statistical structure** is a triple (M, g, C) , where C is a *3-covariant, totally symmetric tensor* on the Riemannian manifold (M, g) , called the skewness tensor.

From a statistical structure (M, g, C) one can construct a **statistical manifold** (M, g, ∇, ∇^*) .

Some Concluding Remarks

The study that began with Fisher-Rao geometry went far beyond its initial statistical scope, and is further expanding in many different new directions

- Fisher-Rao Riemannian geometry
- Amari's Dual connection information geometry
- Optimal transport geometry
- Symplectic geometry, Kähler manifolds and Siegel domains
- Quantum information geometry

References



C.R. Rao (1945)

Information and the accuracy attainable in the estimation of statistical parameters

Bull. Calcutta Math. Soc. 37, 81–89.



S. Amari, Barndorff-Nielsen, Kass, Lauritzen, and C.R. Rao (1987)

Differential Geometry in Statistical Inference

Lecture Notes-Monograph Series. Institute of Mathematical Statistics.



S. Amari (1982)

Differential geometry of curved exponential families - curvature and information loss

Ann. Stat. 10, 375–385



S. Amari and H. Nagaoka (2000)

Methods of Information Geometry

Oxford University Press



Nikolai N. Chentsov (1982)

Statistical decision rules and optimal inference

Monographs, American Mathematical Society

Thank You!